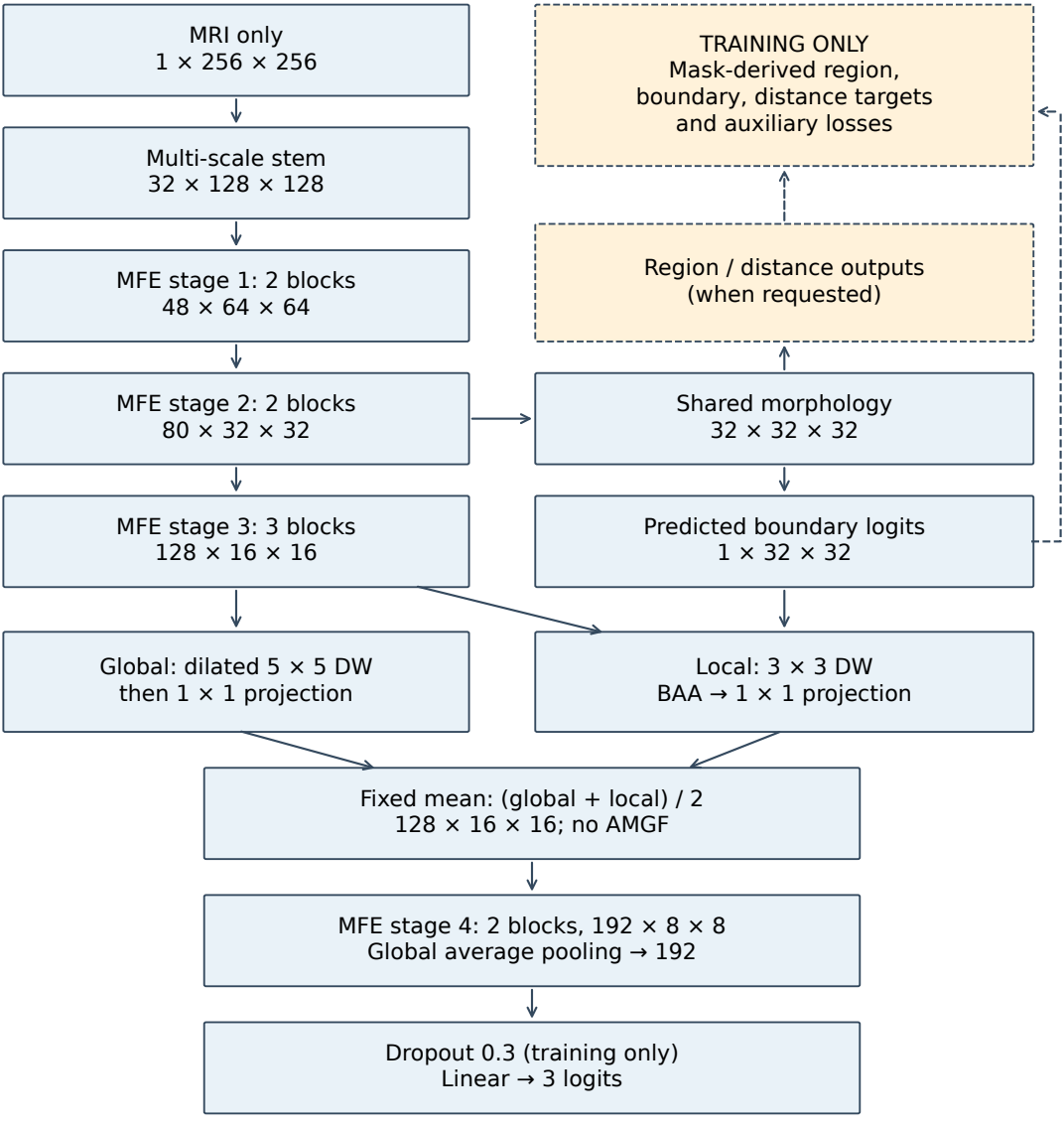
MRI only
1 × 256 × 256
Multi-scale stem
32 × 128 × 128
MFE stage 1: 2 blocks
48 × 64 × 64
MFE stage 2: 2 blocks
80 × 32 × 32
MFE stage 3: 3 blocks
128 × 16 × 16
Global: dilated 5 × 5 DW
then 1 × 1 projection
TRAINING ONLY
Mask-derived region,
boundary, distance targets
and auxiliary losses
Region / distance outputs
(when requested)
Shared morphology
32 × 32 × 32
Predicted boundary logits
1 × 32 × 32
Local: 3 × 3 DW
BAA → 1 × 1 projection
Fixed mean: (global + local) / 2
128 × 16 × 16; no AMGF
MFE stage 4: 2 blocks, 192 × 8 × 8
Global average pooling → 192
Dropout 0.3 (training only)
Linear → 3 logits